

# AYMANE AIT DADS

Machine Learning Intern | Embeddings & Agentic Systems · LLM Fine-Tuning · Multimedia Data Science  
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## PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer - directly demonstrating production-grade LLM fine-tuning at scale. Reduced manual network analysis time by 60% at Orange Maroc by deploying a Random Forest and K-Means pipeline across 100K+ records, translating model outputs into actionable maintenance recommendations for a network optimization team. Brings hands-on expertise in agentic pipelines, RAG, transformer fine-tuning, and embeddings - the exact technical stack of IAS's Multimedia Services Embeddings and Agentic Systems team. Aims to apply LLM orchestration, multimodal embeddings, and ablation-driven experimentation to advance IAS's media measurement and optimisation platform at scale.

## CORE COMPETENCIES

Embeddings & Agentic Systems | LLM Fine-Tuning | Transformer Models | Media Measurement Optimisation | Multimedia Data Science | RAG Pipelines | Ablation Studies | LLM Orchestration

## WORK EXPERIENCE

### Orange Maroc

Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Built an automated feature engineering and model evaluation pipeline using Random Forest and K-Means on **100K+ fiber-optic network records**, classifying quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production use.
- Reduced manual analysis time by **~60%** through end-to-end automation of data ingestion, feature extraction, and model scoring across upload speed, latency, and jitter dimensions.
- Delivered stakeholder presentations translating complex model outputs into concrete maintenance recommendations, using **Power BI** dashboards to communicate findings to non-technical audiences.

## PROJECTS

### NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context, targeting all attention projections, MLP layers, and lm\_head).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce **verified chain-of-thought datasets** with zero boxed-answer mismatches.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

### AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head, dual learning rates, and FP16 mixed precision on 250K images spanning 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; achieved best generalization at 1 epoch (0.793 AUC vs. 0.767 AUC at 4 epochs), validated via a 3-model ensemble with 10-view test-time augmentation.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

## EDUCATION

### Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Image Security, Statistics, Web Applications

### CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

## SKILLS

**ML Frameworks & Engineering:** PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | Scikit-learn | TensorFlow

**LLM & Agentic Systems:** LoRA | QLoRA | SFTTrainer | vLLM | RAG | LangChain | Agentic Pipelines | Claude API | Prompt Engineering

**Data Science & MLOps:** Python | Pandas | NumPy | SQL | Power BI | Docker | Git | BF16/FP16 Mixed Precision | CUDA